

Domain $\rightarrow$ all the x -values the function is allowed to use.

Polynomials: eg: $3x^2 - 2x + 4, \dots$

Domains are all real no: $(-\infty, \infty)$

* no divisions by 0

* no square root of negative

Rational function; $\frac{x+2}{x-5}$,
for domain,

$$x-5 \neq 0, \quad x \neq 5, \quad \text{Domain} = (-\infty, 5) \cup (5, \infty)$$

Square roots (even power): $\sqrt{x+5}, \sqrt[4]{x+6}$

for domain, $x+3 \geq 0$

$$x \geq -3$$

Domain: ~~$(-\infty, -3]$~~ $[-3, \infty)$

Square roots but odd power: $\sqrt[3]{x+3}, \dots$
Domain are real numbers: $(-\infty, \infty)$

Combinations of functions (+, -, $\times$)

$f(x) \times g(x) \rightarrow$ Domain $\rightarrow$ intersection of functions.

$$(1) f(x) = 8x + 3$$

$$g(x) = \sqrt{x}$$

$$f\{g(x)\} = 8(\sqrt{x}) + 3$$

$$= 8\sqrt{x} + 3$$

$$g\{f(x)\} = \sqrt{8x+3}$$

Domain for,

$$(g \circ f)(x) = \sqrt{8x+3}$$

Inside square root must be ≥ 0 :

$$8x + 3 \geq 0$$

$$8x \geq -3$$

$$x \geq -\frac{3}{8}$$

$$\text{Domain; } \left[-\frac{3}{8}, \infty\right)$$

Domain:

$$(f \circ g)(x) = 8\sqrt{x} + 3$$

We need x such that $g(x)$ is defined & $f(g(x))$ is defined.

$\rightarrow g(x) = \sqrt{x}$ requires $x \geq 0$
(Can't take square of $-ve$)

$\rightarrow f\{g(x)\} = 8\sqrt{x} + 3$ is fine
for all $x \geq 0$

* Domain of $f \circ g(x) : [0, \infty)$

#0 Given:

$$x - 2y = 8 \text{ --- (I)}$$

$$x = y + 5 \text{ --- (II)}$$

Using substitution method,

$$x - 2y = 8$$

$$\text{or, } (y+5) - 2y = 8 \text{ [from eq (II)]}$$

$$\text{or, } y + 5 - 2y = 8$$

$$\text{or, } 5 - y = 8$$

$$\text{or, } 5 - 8 = y$$

$$\boxed{y = -3}$$

Put the value of 'y' in equation (I)

$$x = y + 5$$

$$= -3 + 5$$

$$= 2$$

$$\therefore (x, y) = (2, -3)$$

① Domain of the function:

$$f(x) = \frac{x+1}{7-x}$$

* This is rational function

For domain, $7-x \neq 0$

$$7 \neq x$$

$$\therefore x \neq 7$$

So, domain is all real number except 7.

(7 is excluded)

$\therefore$ Domain: $(-\infty, 7) \cup (7, \infty)$

③ $f(x) = \frac{1}{2}x^3 - 5$, find inverse, domain & Range.

$$\text{Let, } y = \frac{1}{2}x^3 - 5$$

$\downarrow$
all 'x' value

$\downarrow$
all 'y' value

Swap/Interchange value/place of x & y,

$$x = \frac{1}{2}y^3 - 5$$

$$x+5 = \frac{1}{2}y^3$$

$$2(x+5) = y^3$$

$$y = \sqrt[3]{2(x+5)}$$

$$f^{-1}(x) = \sqrt[3]{2(x+5)}$$

$$= \sqrt[3]{2x+10}$$

Domain of, $f(x) = \frac{1}{2}x^3 - 5$ [This is cubic polynomial, so, domain are real num]

$$\text{Domain} = (-\infty, \infty)$$

$$\text{Range} = (-\infty, \infty)$$

$$\text{Domain of } f^{-1}(x) = \sqrt[3]{2x+10}$$

[This is cube root -
cube roots accept
ALL real num including
negatives]

$$\text{Domain: } (-\infty, \infty)$$

$$\text{Range: } (-\infty, \infty)$$

⑤ Solve; $\frac{6}{x+7} = \frac{8}{x}$

Cross multiplication Rule

$$\frac{6}{x+7} \rightarrow \frac{8}{x}$$

[Divide $\rightarrow$ multiply]

$$6 \times x = 8(x+7)$$

$$6x = 8x + 56$$

$$-56 = 8x - 6x$$

$$-56 = 2x$$

$$\frac{-56}{2} = x$$

$$\boxed{x = -28}$$

$$\left[\begin{array}{l} \frac{a}{b} = \frac{c}{d} \\ a \times d = c \times b \end{array} \right]$$

⑥ $\sqrt{x+9} + 3 = x$

$$\sqrt{x+9} = x - 3$$

Squaring both sides

$$(\sqrt{x+9})^2 = (x-3)^2$$

$$x+9 = x^2 - 2 \cdot x \cdot 3 + 3^2$$

$$x+9 = x^2 - 6x + 9$$

$$9-9 = x^2 - 6x - x$$

$$0 = x^2 - 7x$$

$$x^2 - 7x = 0$$

$$x(x-7) = 0 \Rightarrow \text{Take common}$$

Either

$$\boxed{x = 0}$$

or

$$\boxed{x - 7 = 0}$$

$$\boxed{x = 7}$$

Checking

when $x=0$

$$\sqrt{x+9} + 3 = x$$

$$\sqrt{0+9} + 3 = 0$$

$$\sqrt{9} + 3 = 0$$

$$\sqrt{3^2} + 3 = 0$$

$$3 + 3 = 0 \quad \text{[false]}$$

when $x=7$

$$\sqrt{x+9} + 3 = x$$

$$\sqrt{7+9} + 3 = 7$$

$$\sqrt{16} + 3 = 7$$

$$\sqrt{4^2} + 3 = 7$$

$$4 + 3 = 7$$

$$7 = 7 \text{ (True)}$$

So, value of

$$\underline{\underline{x = 7}}$$

#



$A(x_1, y_1)$ $B(x_2, y_2)$

Let, A & B are two points;

Distance between A & B point:

A $\longrightarrow$ (x_1, y_1)

B $\longrightarrow$ (x_2, y_2)

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Slope: Any straight line can be written as;

$$y = mx + b$$

$\downarrow$
slope

$\downarrow$
Y-intercept

This is called slope intercept form. $\therefore y = mx + b$

* Slope of line joining the points (x_1, y_1) & (x_2, y_2) is

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

* Also, slope of st. line in the form of $ax + by + c = 0$

$$m = - \frac{\text{Coefficient of } x}{\text{Coefficient of } y}$$

The equation of st. line:-

(i) Slope intercept form; $y = mx + b$

(ii) Point Slope form: $y - y_1 = m(x - x_1)$

(iii) Two points form: ~~y~~ $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$

Also, condition on slope

$\rightarrow$ Parallel 
 $m_1 = m_2$

$\rightarrow$ Perpendicular 
 $m_1 \times m_2 = -1$

$$\text{or, } m_2 = -\frac{1}{m_1}$$

⑦ find slope & y-intercept.

$$f(x) = 3x - 2$$

$$\text{let, } y = 3x - 2$$

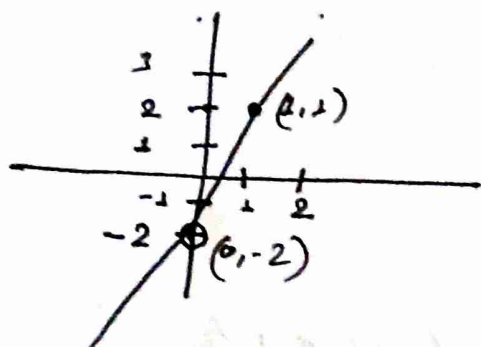
Compare with: $y = mx + b$

$$m(\text{slope}) = 3$$

$$b(\text{y-intercept}) = -2$$

y-intercept in coordinate form; $(0, -2)$

'x' will be always zero (0)
in y-intercept



(st. line,
something like this)

Step 1: point y-intercept

$(0, -2)$

we know,

$$y = \frac{3}{1}x - 2$$

$$m = \frac{3}{1}$$

⑧ Distance; $(-7, -9)$ & $(3, -15)$

(x_1, y_1) (x_2, y_2)

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{\{3 - (-7)\}^2 + \{-15 - (-9)\}^2}$$

$$= \sqrt{(3+7)^2 + (-15+9)^2}$$

$$= \sqrt{10^2 + (-6)^2}$$

$$= \sqrt{100 + 36}$$

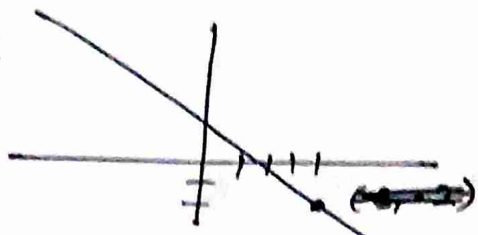
$$= \sqrt{136}$$

$$= 2\sqrt{34}$$

9) Write equation for two lines through point $(3, -2)$:

Parallel to $3x + 4y = 7$ & Perpendicular to $3x + 4y = 7$.

Part 1



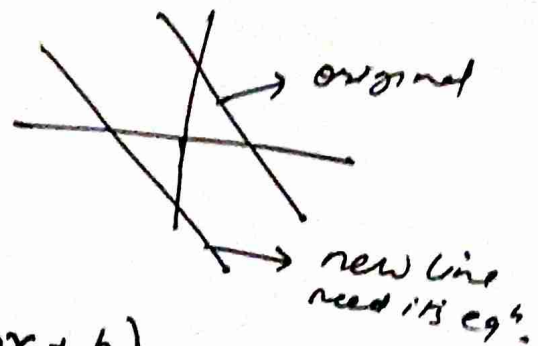
original eqⁿ of line
($3x + 4y = 7$)

Question is asking, eqⁿ of another new line
So, we have: $(3, -2) \rightarrow (x_1, y_1)$
Eqⁿ of line: $3x + 4y = 7$

$$4y = 7 - 3x$$

$$y = \frac{7 - 3x}{4}$$

$$y = -\frac{3}{4}x + \frac{7}{4} \quad (y = mx + b)$$



$$m = -\frac{3}{4}$$

Since, it is parallel; $m_1 = m_2 = -\frac{3}{4}$

Eqⁿ of new line is

$$y - y_1 = m_2(x - x_1)$$

$$y - (-2) = -\frac{3}{4}(x - 3)$$

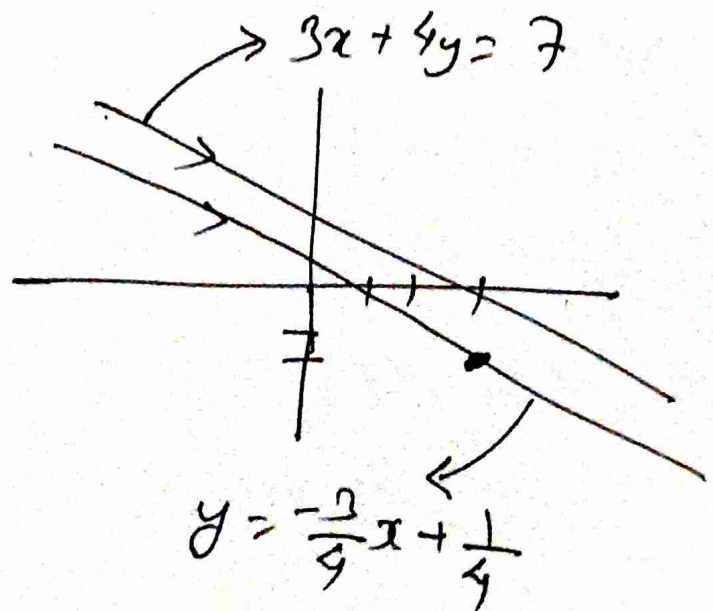
$$y + 2 = -\frac{3}{4}(x - 3)$$

$$4y + 8 = -3x + 9$$

$$3x + 4y =$$

$$y = \frac{-3x + 9 - 8}{4}$$

$$y = -\frac{3}{4}x + \frac{1}{4}$$



Part 2: for perpendicular;

$$3x + 4y = 7$$

$$y = \frac{7-3x}{4}$$

$$y = -\frac{3}{4}x + \frac{7}{4} \quad [y = mx + b]$$

$$m_1 = -\frac{3}{4}$$

Since, it is perpendicular

$$m_2 = -\frac{1}{m_1} = \frac{-1}{-\frac{3}{4}} = \frac{4}{3}$$

Eqⁿ of new line,

$$y - y_1 = m_2(x - x_1)$$

$$y - (-2) = \frac{4}{3}(x - 3)$$

$$3(y + 2) = 4x - 12$$

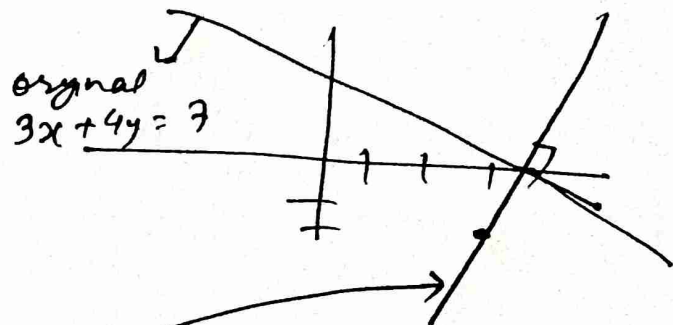
$$3y + 6 = 4x - 12$$

$$3y = 4x - 12 - 6$$

$$y = \frac{4x - 18}{3}$$

$$y = \frac{4}{3}x - 6$$

Passing point
 $(x_1, y_1) \rightarrow (3, -2)$



→ Eqⁿ of st. line
which is
perpendicular